Deep Dive: Quantizing LLMs

Companion videos: https://youtu.be/kw7S-3s50uk + https://youtu.be/fXBBwCIA0Ds

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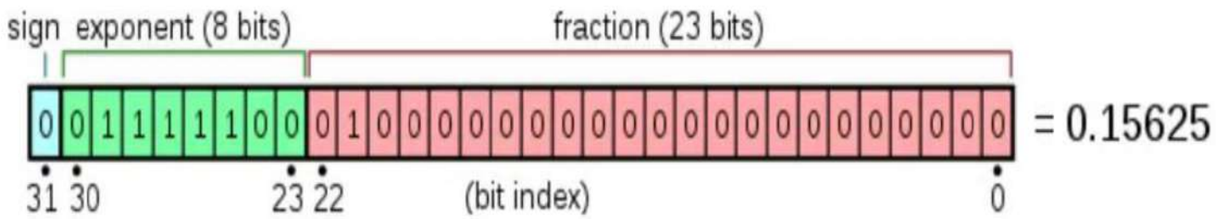
What is quantization?

- Model weights are learned during training, and stored as numerical values
- Common data types: FP32, FP16, BF16



- The larger the data type, the finer the granularity: models are more accurate
- This comes at the cost of higher memory and compute requirements: models are larger, inference is slower
- The purpose of quantization is to rescale weights and/or activations to shorter data types, to reduce memory and compute requirements while minimizing loss of accuracy
- Common lower-precision data types: INT8, INT4

How can we best map a high-precision data format to a lower-precision format?

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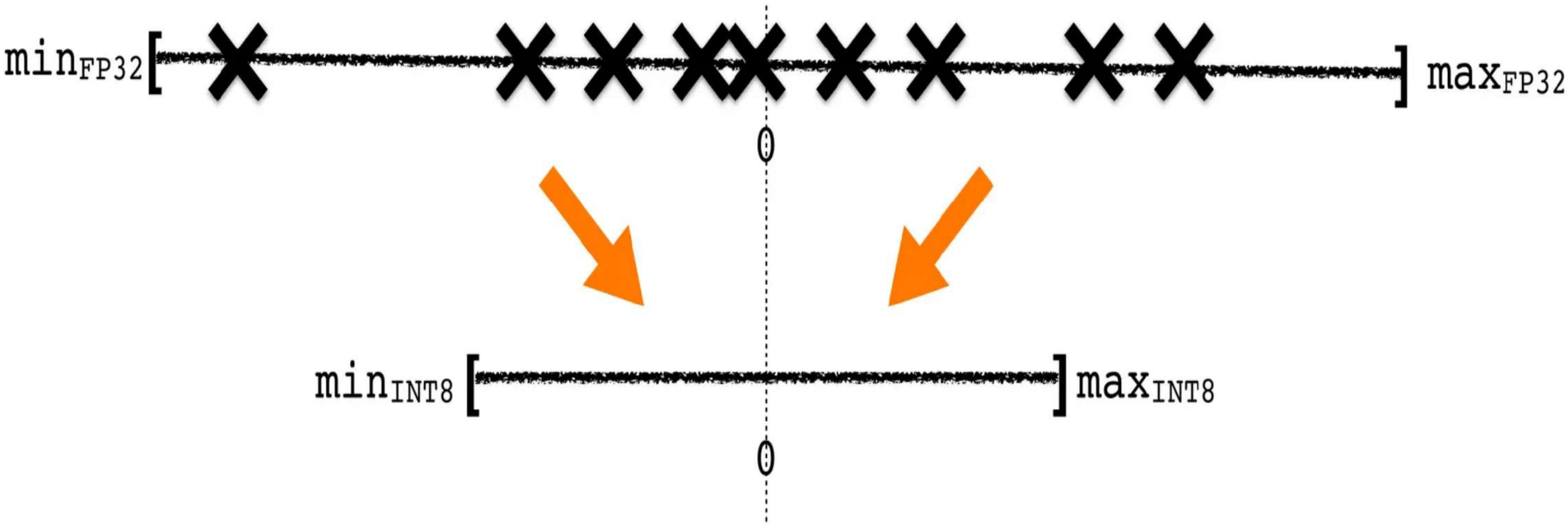
Rescaling weights and activations



Note: only one dimension is shown,
but we need to do the same on all tensor dimensions!

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Mapping function

The mapping function maps the high-precision numerical space to a lower-precision space

This usually takes place per layer, but finer-grained quantization is also possible (e.g. per row or per column)

$$Q(r) = round\left(\frac{r}{S} + Z\right)$$

S,Z: quantization parameters

$$S = \frac{\beta - \alpha}{\beta_q - \alpha_q}$$

Ratio of the $[\alpha,\beta]$ input range to the $[\alpha_q,\beta_q]$ output range

For 8-bit quantization: $\beta_q - \alpha_q = 2^8 - 1$

$$Z = \alpha_q - \frac{\alpha}{S}$$

Bias value to map a zero input to a zero output

How can we best pick the $[\alpha,\beta]$ range to minimize quantization error?

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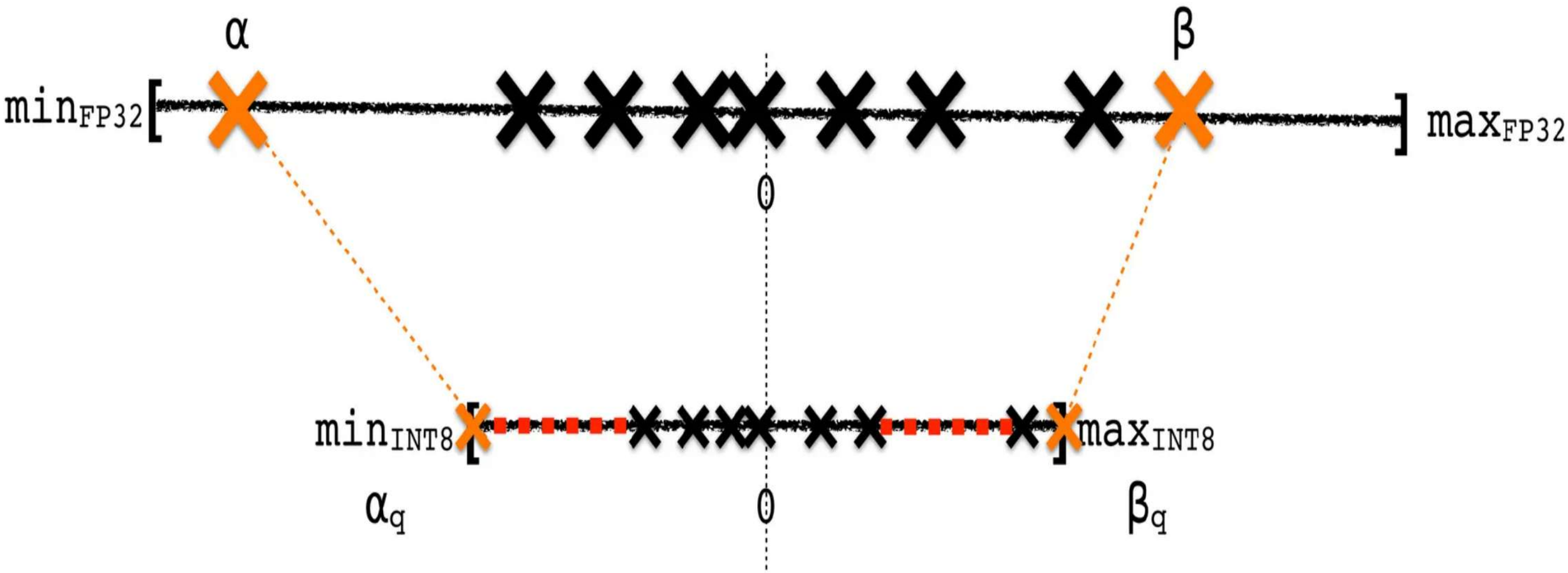
Picking the [α,β] range

	Range	Min. positive value
FP32 / BF16	[±1.18e-38, ±3.4e38]	1.4e-45
INT8	[-128, +127]	1



A simple technique is to use the minimum and maximum value of the input range

This is sensitive to outliers: part of $[\alpha_q,\beta_q]$ could be unused, and "squeeze" other values

Alternative: use percentiles or histogram bins

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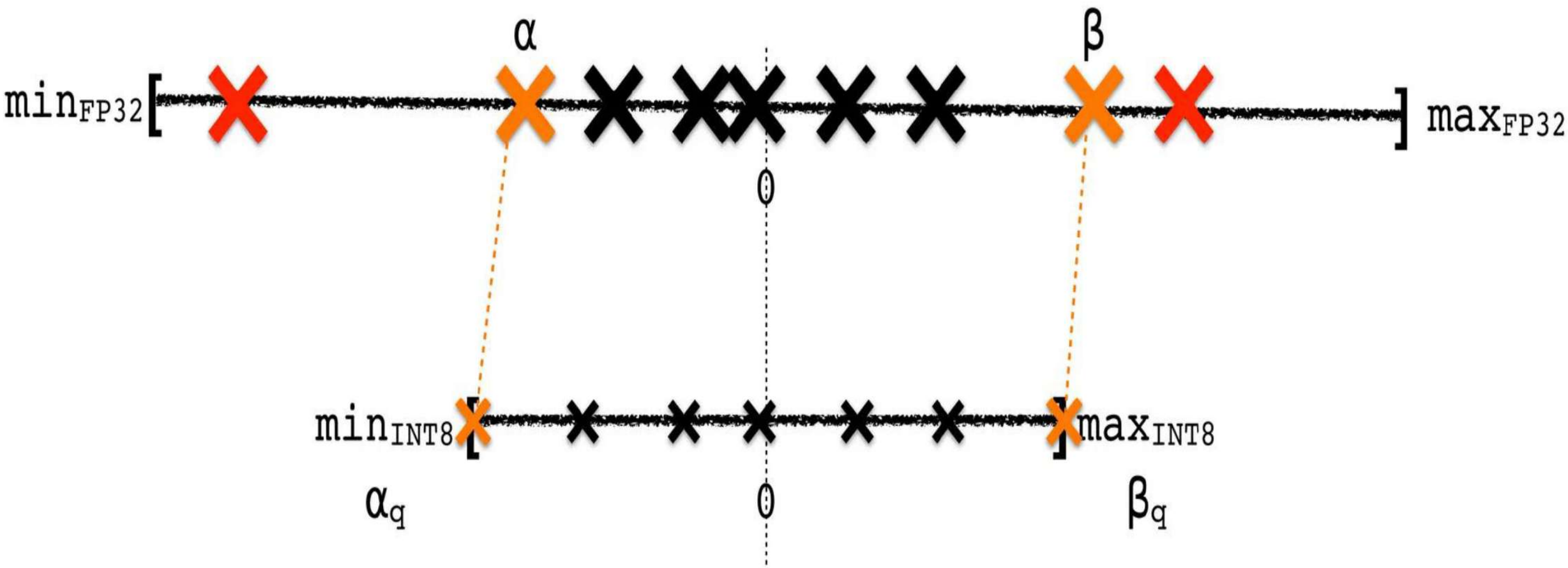

To make the best use of $[\alpha_q,\beta_q]$, we need to get rid of outliers

How many can we eliminate without hurting accuracy too much?

We need to minimize information loss between the two distributions

➡️ Use a calibration dataset and minimize the Kullback-Leibler (KL) divergence 😱

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When can we apply quantization?

- Post-training dynamic quantization (dynamic PTQ)
  - Load the trained model for inference and convert model weights ahead of time
  - Rescale activations on the fly, just before running the computation
  - The scale factor is computed dynamically 👍 simple, more flexibility 👎 some overhead
- Post-training static quantization (static PTQ)
  - Load the trained model for inference and convert model weights ahead of time
  - Predict a calibration dataset, and use the observed distribution of activations to set the scale factor
  - 👍 no overhead, lower latency 👎 calibration may not work well for all datasets, need different models
- Quantization-aware training (QAT)
  - Training still takes place with full precision
  - Weights and activations are "fake quantized" during training
  - 👍 higher quality models 👎 need to retrain, expensive for LLMs

- At inference time, a dequantization process returns results in the original data type

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Dynamic PTQ with PyTorch

https://pytorch.org/tutorials/intermediate/dynamic_quantization_bert_tutorial.html

```
import torch
from transformers import BertTokenizer, BertModel

device = torch.device("cuda:0" if torch.cuda.is_available() else "cpu")

tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
orig_model = BertModel.from_pretrained("bert-base-uncased").to(device=device)

quantized_model = torch.quantization.quantize_dynamic(
    orig_model, {torch.nn.Linear}, dtype=torch.qint8
)
```

Quantization of the Linear layers to INT8

Model size: 437MB ➡️ 181MB (-58%)

Inference latency (1 thread): 34.7 ms ➡️ 24 ms (-31%)

<1% F1 degradation on MRPC task

```
(output): BertOutput(
          (dense): DynamicQuantizedLinear(in_features=3072,
out_features=768, dtype=torch.qint8, qscheme=torch.per_tensor_affine)
          (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (dropout): Dropout(p=0.1, inplace=False)
        )
```

Amazon EC2 c6i.4xlarge, AWS Deep Learning AMI, PyTorch 2.2.0 + IPEX 2.2.0

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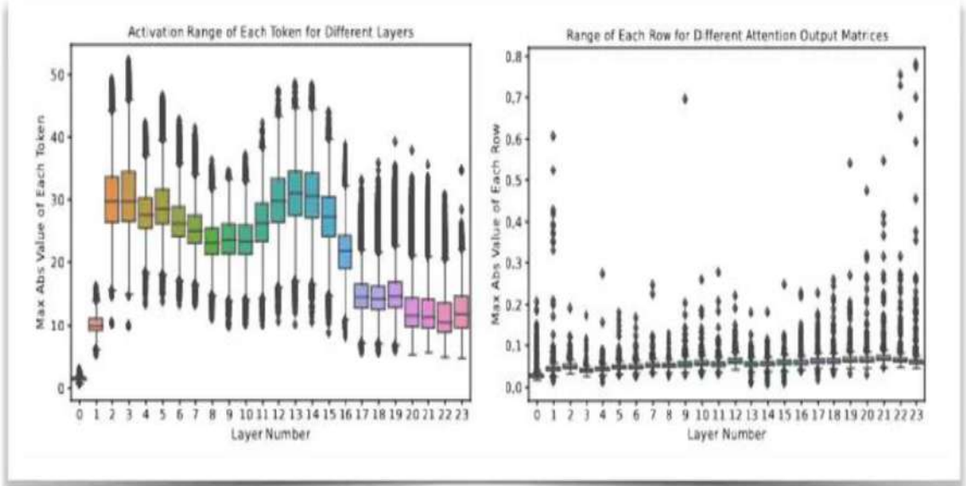
ZeroQuant

https://arxiv.org/abs/2206.01861 (06/2022) + https://github.com/mit-han-lab/smoothquant

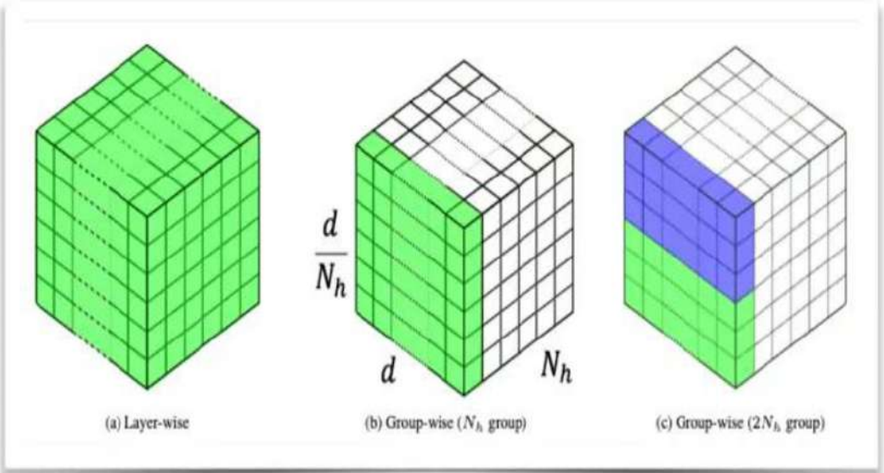
- First (?) dynamic post-training quantization optimized for LLMs
- INT8 weights and activations, or INT4 weights and INT8 activations
- Weight ranges are very different across layers, activation ranges even more so
- Group-wise quantization for weights https://arxiv.org/abs/1909.05840 (09/2019) partition tensors into weight groups, and quantize them separately
- Token-wise quantization for activations: dynamically compute the min/max range for each token
- Layer-by-layer knowledge distillation (LKD) for increased accuracy (optional)
- Optimized implementations for GPU hardware
- 2x-5x speedup (BERT), negligible accuracy drop for W8A8
- Stronger quantization has a larger impact, particularly on generative tasks

Precision (Method)	Lambada (↑)	PIQA (↑)	OpenBookQA (↑)	RTE (↑)	ReCoRd (↑)	Ave. 19 Tasks (↑)	Wikitext-2 (↓)	Time Cost
W16A16	61.3	71.4	33.6	53.1	82.6	42.4	15.3	N/A
W8A8 (PTQ)	54.8	67.7	16.6	54.5	75.7	40.5	18.9	13 mins
W8A8 (ZeroQuant)	62.6	70.7	33.4	52.7	80.9	42.3	15.7	0
W4/8A16 (PTQ)	0.00	50.4	27.0	50.9	15.8	29.0	1.35e5	13 mins
W4/8A16 (ZeroQuant)	43.9	66.5	30.0	52.7	77.3	39.38	21.9	0
W4/8A16 (ZeroQuant-LKD)	59.4	69.5	31.6	52.7	79.7	41.5	17.6	3 hours
W4/8A8 (ZeroQuant)	46.8	66.4	28.8	52.7	76.2	39.24	24.1	0
W4/8A8 (ZeroQuant-LKD)	48.7	68.1	29.0	52.0	77.4	39.90	18.2	3 hours

GPT-3 1.3B



The activation range (left) and row-wise weight range of the attention output matrix (right) of different layers of GPT-3 350M



Group-wise quantization

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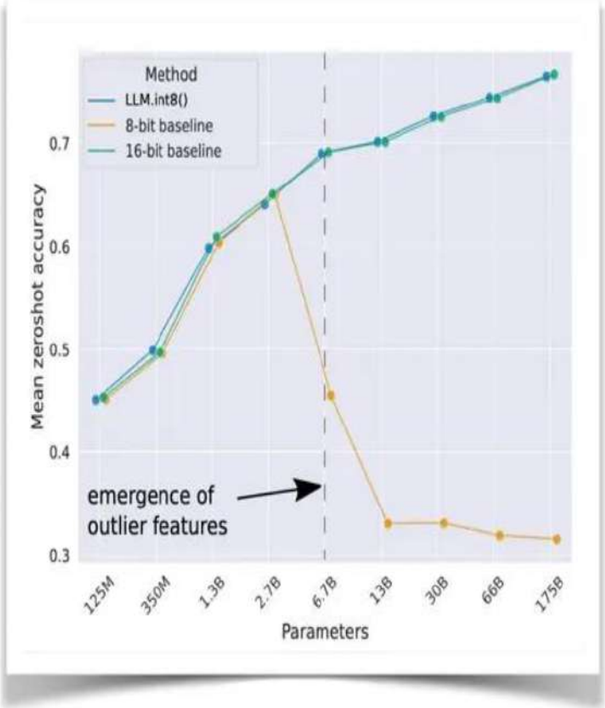
bitsandbytes

https://arxiv.org/abs/2208.07339 (11/2022) + https://huggingface.co/blog/hf-bitsandbytes-integration

- Dynamic post-training quantization
- Quantization primitives for INT8/INT4 linear layers (weights only), and INT8 optimizers
- "*We find that outlier features strongly affect attention and the overall predictive performance of transformers. While up to 150k outliers exist per 2048 token sequence for a 13B model, these outlier features are highly systematic and only representing at most 7 unique feature dimensions.*"
- Vector-wise quantization: use different scaling factors when multiplying input and weight tensors
- Outlier dimensions are quantized separately with mixed-precision decomposition (16-bit)
- 8-bit quantization vs FP16: 2x memory savings, accuracy on par, can be faster for LLMs
- Mixed-precision decomposition is difficult to implement efficiently on hardware accelerators
- Integrated in `transformers`, `accelerate`, `peft`, and TGI



Parameters	125M	1.3B	2.7B	6.7B	13B
32-bit Float	25.65	15.91	14.43	13.30	12.45
Int8 absmax	87.76	16.55	15.11	14.59	19.08
Int8 zeropoint	56.66	16.24	14.76	13.49	13.94
Int8 absmax row-wise	30.93	17.08	15.24	14.13	16.49
Int8 absmax vector-wise	35.84	16.82	14.98	14.13	16.48
Int8 zeropoint vector-wise	25.72	15.94	14.36	13.38	13.47
Int8 absmax row-wise + decomposition	30.76	16.19	14.65	13.25	12.46
Absmax LLM.int8() (vector-wise + decomp)	25.83	15.93	14.44	**13.24**	**12.45**
Zeropoint LLM.int8() (vector-wise + decomp)	**25.69**	**15.92**	**14.43**	**13.24**	**12.45**

C4 validation perplexities on OPT models (lower is better)

GPT-3 Size Model dimension	Small 768	Medium 1024	Large 1536	XL 2048	2.7B 2560	6.7B 4096	13B 5140	175B 12288
FP16-bit baseline	1.00x	1.00x	1.00x	1.00x	1.00x	1.00x	1.00x	1.00x
Int8 without overhead	0.99x	1.08x	1.43x	1.61x	1.63x	1.67x	2.13x	2.29x
Absmax PyTorch+NVIDIA	0.25x	0.24x	0.36x	0.45x	0.53x	0.70x	0.96x	1.50x
Vector-wise PyTorch+NVIDIA	0.21x	0.22x	0.33x	0.41x	0.50x	0.65x	0.91x	1.50x
Vector-wise	**0.43x**	**0.49x**	**0.74x**	**0.91x**	**0.94x**	**1.18x**	**1.59x**	**2.00x**
LLM.int8() (vector-wise+decomp)	0.14x	0.20x	0.36x	0.51x	0.64x	0.86x	1.22x	1.81x

Inference speedup (higher is better)

```
from transformers import AutoModelForCausalLM

model_8bit = AutoModelForCausalLM.from_pretrained(
    "bigscience/bloom-1b7", device_map="auto", load_in_8bit=True)
```

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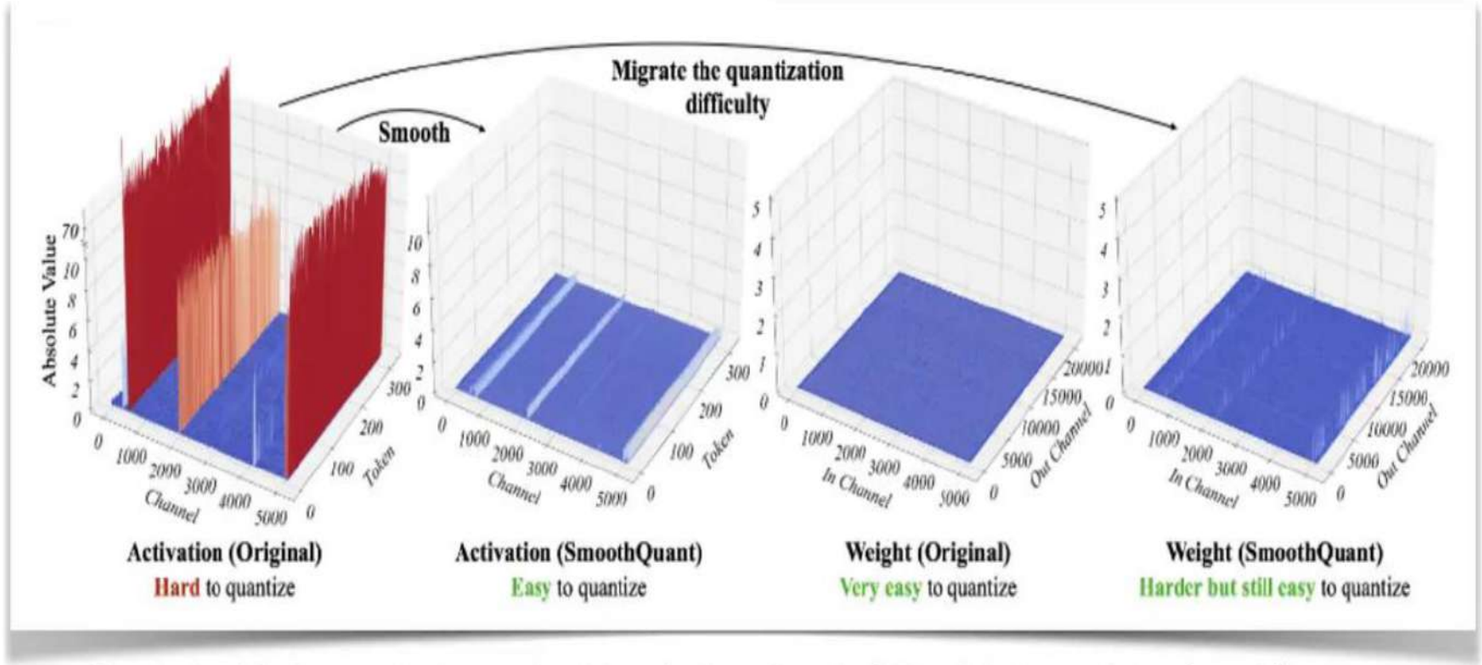
SmoothQuant

https://arxiv.org/abs/2211.10438 (11/2022) + https://github.com/mit-han-lab/smoothquant

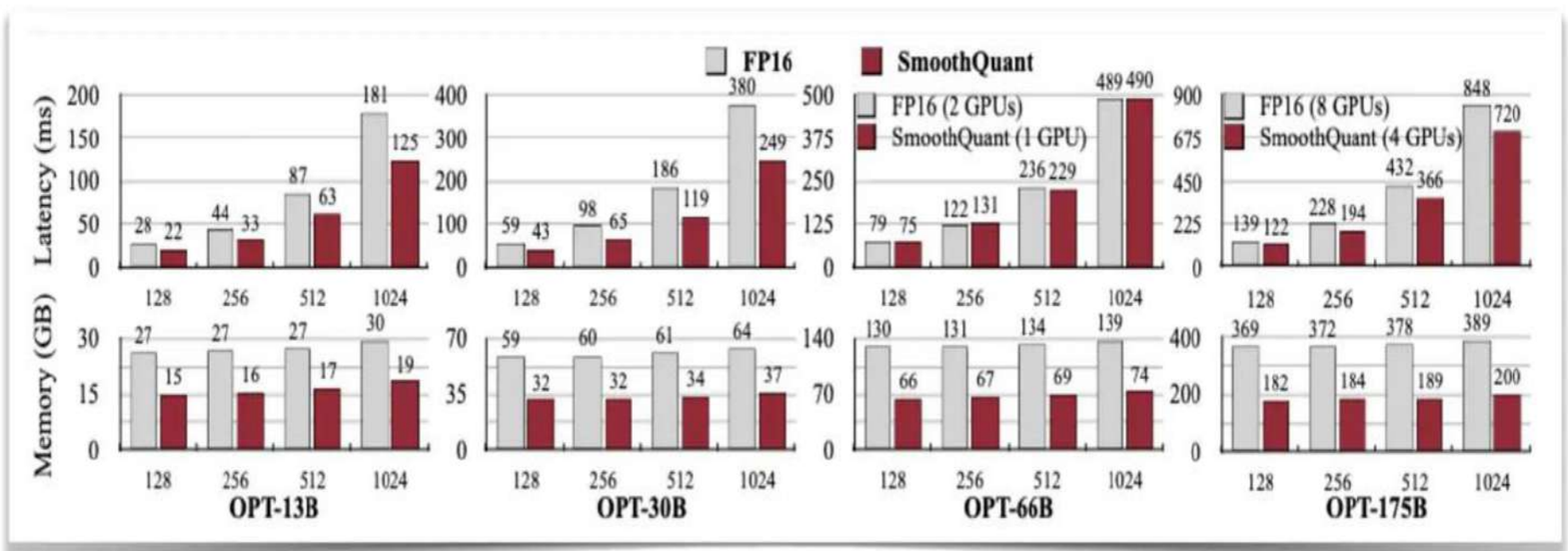
- Dynamic or static post-training quantization
- Activations are harder to quantize (much larger range)
- Migrate the quantization difficulty to the weight layers
- Rescale weights to reduce the magnitude of the activation ranges
  - Rescale activations of self-attention and linear layers
  - Quantize linear layers to INT8
- As accurate as bitsandbytes, much faster



Magnitude of the input activations and weights of a linear layer in OPT-13B before and after SmoothQuant.

Method	OPT-175B	BLOOM-176B	GLM-130B*
FP16	71.6%	68.2%	73.8%
W8A8	32.3%	64.2%	26.9%
ZeroQuant	31.7%	67.4%	26.7%
`LLM.int8()`	71.4%	68.0%	73.8%
Outlier Suppression	31.7%	54.1%	63.5%
SmoothQuant-O1	**71.2%**	68.3%	**73.7%**
SmoothQuant-O2	71.1%	**68.4%**	72.5%
SmoothQuant-O3	71.1%	67.4%	72.8%

Average accuracy
on WinoGrande, HellaSwag, PIQA, and LAMBADA



Inference latency (top) and memory usage (bottom) of the FasterTransformer implementation on NVIDIA A100-80GB GPUs

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Group-wise Precision Tuning Quantization (GPTQ)

https://arxiv.org/abs/2210.17323 (03/2023) + https://github.com/AutoGPTQ/AutoGPTQ
https://huggingface.co/blog/gptq-integration + https://huggingface.co/blog/overview-quantization-transformers

- Static post-training quantization
- New levels of quantization: 8, 4, 3, or 2-bit precision (weights only)
- Row-wise quantization (aka per-channel), plus modified Optimal Brain Quantization https://arxiv.org/abs/2208.11580 (08/2022)
- 4-bit quantization vs. FP16: 4x memory savings, same inference speed, negligible perf degradation
- The `exllamav2` kernels are enabled by default for faster inference https://github.com/turboderp/exllamav2
- Integrated in `transformers`, `accelerate`, `peft`, and TGI

Method	Bits	OPT-175B				BLOOM-176B			
		Wiki2	PTB	C4	LAMB. ↑	Wiki2	PTB	C4	LAMB. ↑
Baseline	16	8.34	12.01	10.13	75.59	8.11	14.59	11.71	67.40
RTN	4	10.54	14.22	11.61	71.34	8.37	15.00	12.04	66.70
GPTQ	4	**8.37**	**12.26**	**10.28**	**76.80**	**8.21**	**14.75**	**11.81**	**67.71**
RTN	3	7.3e3	8.0e3	4.6e3	0	571.	107.	598.	0.17
GPTQ	3	**8.68**	**12.68**	**10.67**	**76.19**	**8.64**	**15.57**	**12.27**	**65.10**
GPTQ	3/g1024	8.45	12.48	10.47	77.39	8.35	15.01	11.98	67.47
GPTQ	3/g128	8.45	12.37	10.36	76.42	8.26	14.89	11.85	67.86

```
from transformers import AutoModelForCausalLM, AutoTokenizer, GPTQConfig

model_id = "facebook/opt-125m"
tokenizer = AutoTokenizer.from_pretrained(model_id)

quantization_config = GPTQConfig(bits=4, dataset="c4", tokenizer=tokenizer)

# Quantize model
model = AutoModelForCausalLM.from_pretrained(model_id,
device_map="auto",quantization_config=quantization_config)
```

INT3 OPT-175B
average per-token latency
batch size 1, sequence length 128

GPU	FP16	3bit	Speedup	GPU reduction
A6000 – 48GB	589ms	130ms	4.53×	8 → 2
A100 – 80GB	230ms	71ms	3.24×	5 → 1

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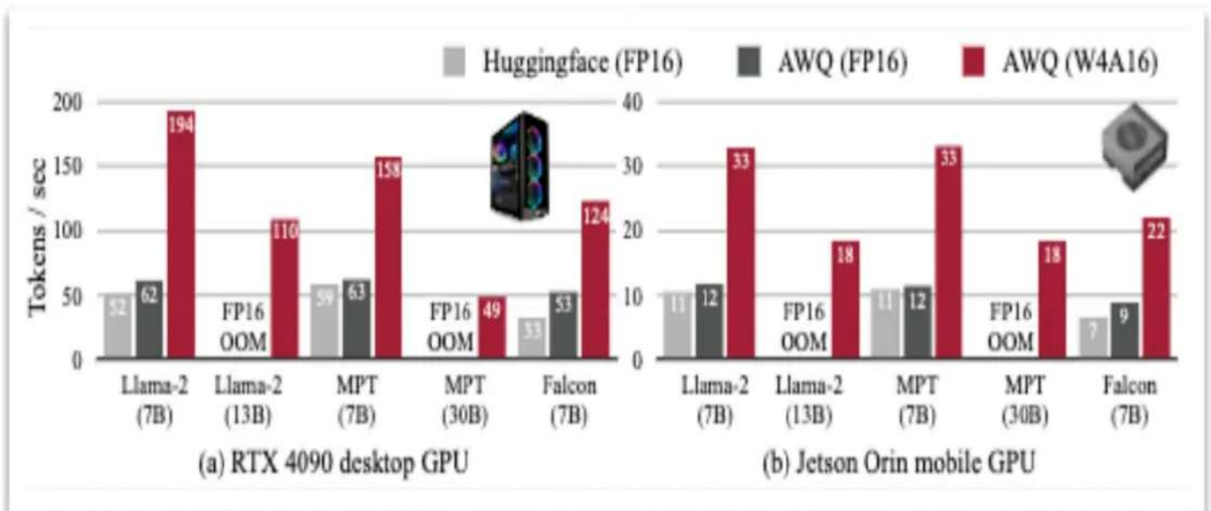
Activation-aware Weight Quantization (AWQ)

https://arxiv.org/abs/2306.00978 (10/2023)

- Static post-training quantization
- 0.1-1% of weights have a critical influence: they are not quantized to minimize degradation
- These salient weights are identified by looking at activations
- AWQ needs 10x less calibration data than GPTQ, and is more robust across datasets
- AWQ performs well on instruction-tuned and multimodal models
- At the moment, `transformers` supports model-loading only
- Models can be quantized with https://github.com/mit-han-lab/llm-awq or https://github.com/casper-hansen/AutoAWQ

PPL↓		Llama-2			LLaMA			
		7B	13B	70B	7B	13B	30B	65B
FP16	-	5.47	4.88	3.32	5.68	5.09	4.10	3.53
INT3 g128	RTN	6.66	5.52	3.98	7.01	5.88	4.88	4.24
	GPTQ	6.43	5.48	3.88	8.81	5.66	4.88	4.17
	GPTQ-R	6.42	5.41	3.86	6.53	5.64	4.74	4.21
	AWQ	**6.24**	**5.32**	**3.74**	**6.35**	**5.52**	**4.61**	**3.95**
INT4 g128	RTN	5.73	4.98	3.46	5.96	5.25	4.23	3.67
	GPTQ	5.69	4.98	3.42	6.22	5.23	4.24	3.66
	GPTQ-R	5.63	4.99	3.43	5.83	5.20	4.22	3.66
	AWQ	**5.60**	**4.97**	**3.41**	**5.78**	**5.19**	**4.21**	**3.62**

Perplexity (lower is better)



AWQ is up to 3.9× than the FP16 implementation on an NVIDIA 4090 GPU

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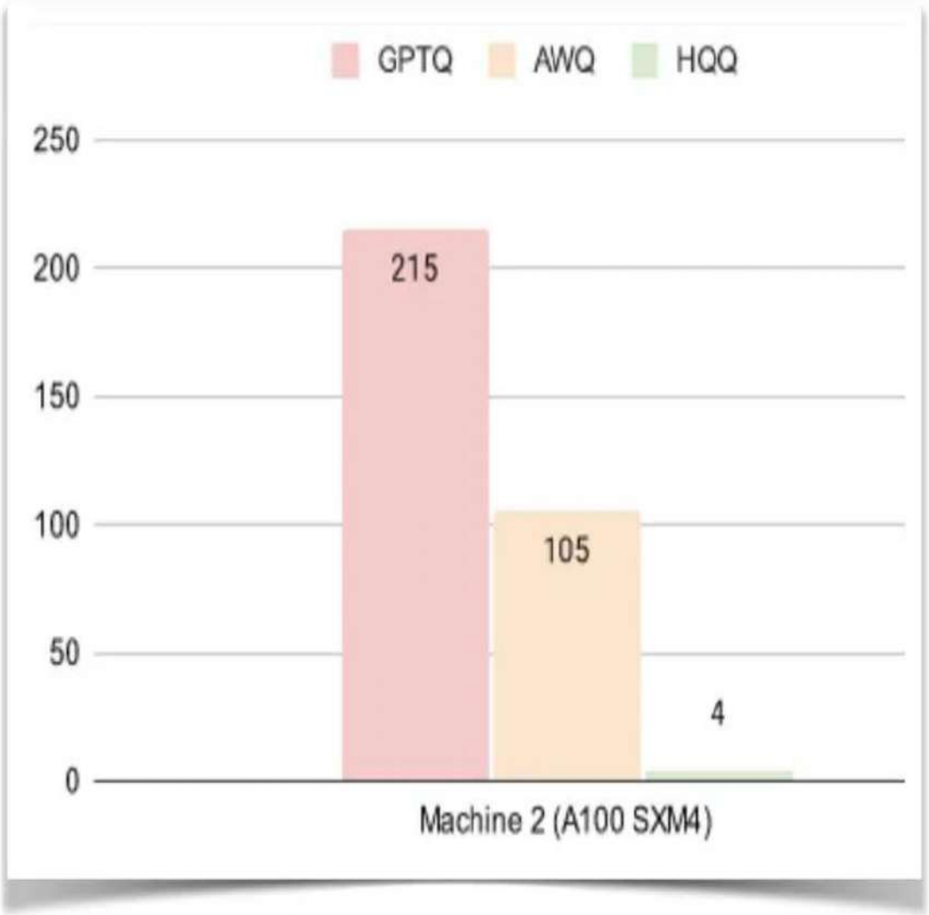
Half-quadratic Quantization (HQQ)

https://mobiusml.github.io/hqq_blog/ + https://github.com/mobiusml/hqq + https://huggingface.co/mobiuslabsgmbh (12/2023)

- Dynamic post-training quantization
- Same accuracy as static PTQ, without the calibration phase
- Minimize error in weight quantization and model outliers
- Outperforms GPTQ and AWQ on most tests



Llama2-70B quantization time (minutes)

Method	nBits	Llama-2-7B		Llama-2-13B		Llama-2-70B	
		PPL ↓	MEM ↓	PPL ↓	MEM ↓	PPL ↓	MEM ↓
FP	16	5.18	13.5	**4.63**	25.6	OOM	OOM
BNB	8	5.22	7.9	4.67	14.4	3.17	**68.15**
GPTQ_g128	8	**5.19**	7.8	**4.63**	14.8	**3.12**	74.87
HQQ_g128	8	**5.19**	**7.6**	**4.63**	**14**	**3.12**	69.32
BNB_g64	4	5.43	4.7	4.79	8.2	3.29	39.11
GPTQ_g128	4	5.41	5	4.74	8.9	3.24	40
GPTQ_g64	4	5.38	5	4.73	9.1	3.23	41.13
AWQ_g128	4	5.32	**4.6**	4.71	8.2	3.21	**35.78**
AWQ_g64	4	**5.28**	**4.6**	**4.7**	8.5	3.2	37.08
HQQ_g128	4	5.35	**4.6**	4.74	**7.9**	3.21	35.97
HQQ_g64	4	5.3	**4.6**	**4.7**	8.2	**3.19**	37.52
GPTQ_g128	3	6.3	3.9	5.25	7	3.85	33.7
GPTQ_g64	3	6.1	4	5.16	7.3	3.7	33.47
HQQ_g128	3	6.2	**3.8**	5.15	**6.8**	3.58	**30.11**
HQQ_g64	3	**5.82**	4.5	**4.98**	7.4	**3.45**	33.46
GPTQ_g64	2	nan	**3.5**	13	6	9.44	24.5
HQQ_g32	2	15.61	**3.5**	7.63	**5.9**	4.82	**24.2**
HQQ_g16	2	**7.3**	4.1	**6.36**	6.9	**4.12**	30.27
HQQ_g16_s*	2	7.31	3.7	6.37	6.1	4.13	26.37

Perplexity on wikitext-2 and memory usage (lower is better)

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Optimum Intel

https://github.com/huggingface/optimum-intel + https://huggingface.co/docs/optimum
https://github.com/huggingface/optimum-intel/tree/main/examples

- Accelerate Hugging Face models on Intel architectures
- Simple transformers-like API and CLI for:
  - Intel Neural Compressor https://github.com/intel/neural-compressor
  - Intel OpenVINO https://github.com/openvinotoolkit/openvino
- Static and dynamic post-training quantization, quantization-aware training
- Different recipes are supported, including SmoothQuant

```
from optimum.intel import INCQuantizer
from neural_compressor.config import AccuracyCriterion, TuningCriterion, PostTrainingQuantConfig

# Set the accepted accuracy loss to 5%
accuracy_criterion = AccuracyCriterion(tolerable_loss=0.05)
# Set the maximum number of trials to 10
tuning_criterion = TuningCriterion(max_trials=10)
quantization_config = PostTrainingQuantConfig(
    approach="dynamic", accuracy_criterion=accuracy_criterion, tuning_criterion=tuning_criterion
)
quantizer = INCQuantizer.from_pretrained(model, eval_fn=eval_fn)
quantizer.quantize(quantization_config=quantization_config,
save_directory="dynamic_quantization")
```

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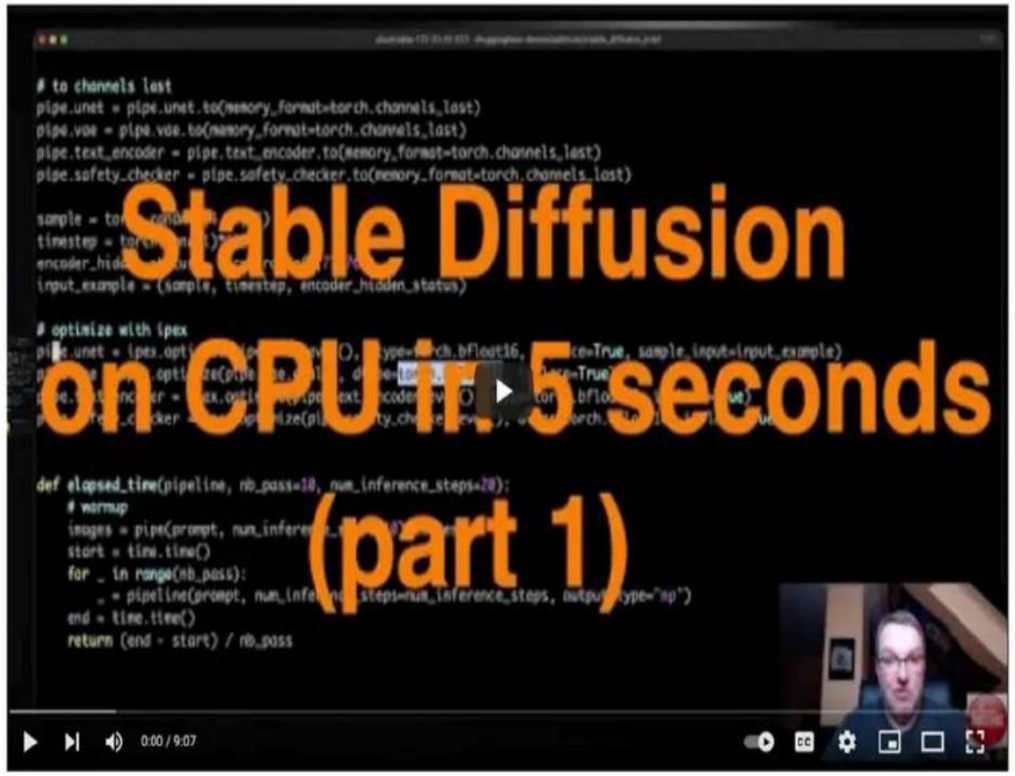
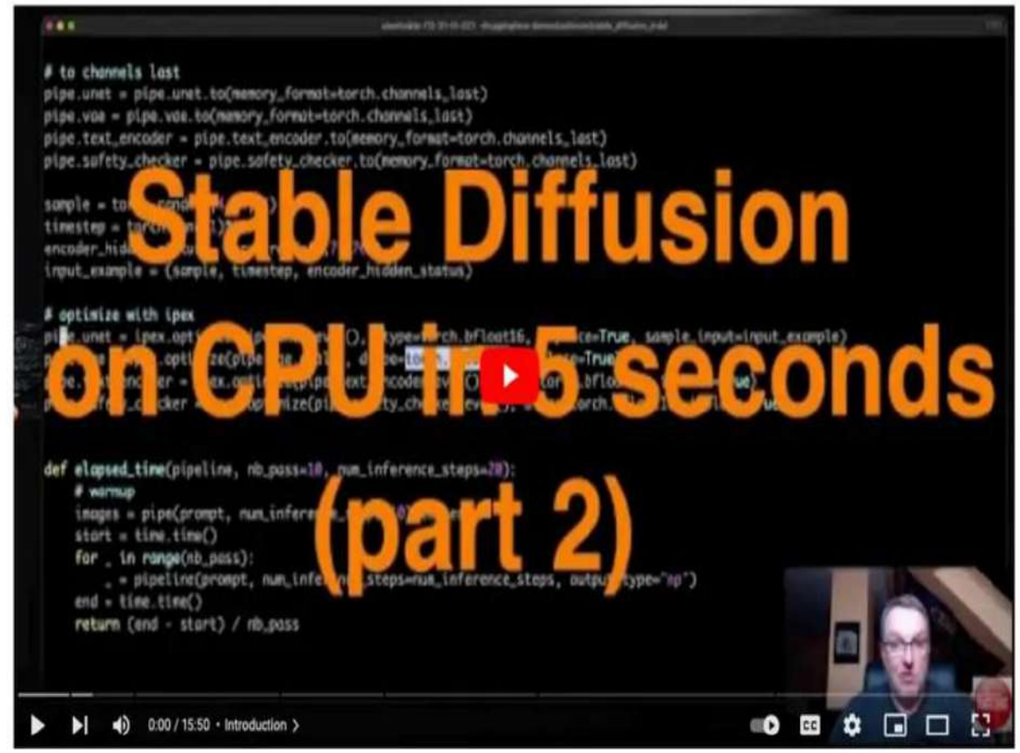
CPU

Demo: Accelerating Stable Diffusion Inference with Intel OpenVINO, IPEX and Hugging Face





https://www.youtube.com/watch?v=KJDCGyZ2fPw

https://www.youtube.com/watch?v=78gE0CHxDbo

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